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|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 2(a)(i)  | <p>any <b>three</b> from:</p> <ol style="list-style-type: none"> <li>1 length of transect ;</li> <li>2 sampling intervals / AW (on transect) ;</li> <li>3 <i>idea of</i> orientation / aspect / AW of, line / transect ;</li> <li>4 time of year / season / AW, (of sampling) ;</li> <li>5 number of sampling points on each line / transect ;</li> <li>6 size of sampling area / quadrat, (at the sampling point) ;</li> <li>7 number of (line) transects ;</li> <li>8 location of transect (in study area) / AW ;</li> </ol> | <b>3</b> |
| 2(a)(ii) | <i>idea of</i> use of, random numbers / coordinates ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>1</b> |
| 2(b)     | <p>(<i>identify all the plant species present and</i>)</p> <p>count, number of individuals / population, of each (plant) species (n) ;</p>                                                                                                                                                                                                                                                                                                                                                                                     | <b>1</b> |
| 2(c)     | 185.7 / 186 ( %) ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>1</b> |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Marks |
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| 2(d)     | <p>any <b>four</b> from:</p> <p>max 3 for supports <b>and</b> p value statement supports</p> <ol style="list-style-type: none"> <li>1 <i>idea that</i> there is a, correlation / relationship, between increase in red quinine ground cover and loss of ground cover of other species (1998 – 2003/5) ;</li> <li>2 <i>idea of</i> as ground cover of red quinine increases, (ground cover of) other species decreases (1998 – 2003/5) ;</li> <li>3 as cover of red quinine increases, cover of, bare ground / no plants, increases (1998 – 2003/5) ;</li> <li>4 there is a, large / AW, decrease in, biodiversity / species diversity, between 1998 and, 2003/5, (when red quinine cover increased) ;</li> <li>5 there is, little difference / AW, in biodiversity / species diversity, between 2003 and 2005 (when quinine cover does not change) ;</li> <li>6 correct data quote to support argument ;</li> </ol> <p><i>correct use of p value award unlinked to support / not support</i></p> <ol style="list-style-type: none"> <li>7. correct use of <math>p = &lt; 0.001</math> with regards to significance (1998 – 2003) ;</li> <li>8. correct use of <math>p = &gt; 0.05</math> with regards to significance (2003 – 2005) ;</li> </ol> <p><i>at least 1 of:</i><br/><i>does not support</i></p> <ol style="list-style-type: none"> <li>9 no causal relationship demonstrated ;</li> <li>10 no data on animal species diversity ;</li> <li>11 no data on ecosystem / habitat / genetic biodiversity ;</li> </ol> | 4     |